Arithmetic and Geometric Sequences – Lesson #2

MCR3U

Sequences Questions

What is the difference between a sequence and a series?

What is the difference between Arithmetic and Geometric?

Formulas for general terms of a sequence

Example 1: $-10, -4, 2, \ldots$

a) Determine whether the sequence is arithmetic or geometric.

b) Determine an equation for the sequence.

c) Find the value of t_{21}

Example 2: Insert two numbers between 8 and 32 so the four numbers form an arithmetic sequence.

Example 3: An arithmetic sequence is 8, 14, 20, 26, ... Which term has the value 92? Prove mathematically.

Example 4: $200, -100, 50, \ldots$

a) Is the sequence arithmetic or geometric?

b) Find an equation to represent the sequence.

c) Find t_{14}

Example 5: Complete the geometric sequence

____, 160, ____, ____, ____, 10

Example 6: The 50th term of an arithmetic sequence is 238 and the 93rd term is 539. Find a general equation to represent the sequence.

Example 7: Determine the number of terms in the geometric sequence: 5, -10, 20, ... , -10 240